

Access this article online

Quick Response Code:



Website:

http://journals.lww.com/TJOP

DOI:

10.4103/tjo.TJO-D-24-00131

Surgical treatment of uveal melanoma: Exoresection versus endoresection

Seung Min Lee¹, Min Kim^{2*}

Abstract:

Uveal melanoma, the most common primary intraocular malignancy in adults, presents significant therapeutic challenges. With the advancement of surgical instruments and techniques, exoresection (ab-externo) and endoresection (ab-interno) have emerged as viable treatment methods. These surgical interventions can be implemented particularly for tumors exceeding traditional size criteria for plaque brachytherapy and for tumors that are too large, which pose an elevated risk of radiation-related complications. Moreover, surgical interventions not only enhance the potential for functional vision preservation and globe retention but also provide effective control of intraocular tumors and metastatic disease when combined with adjunctive radiation therapy. Exoresection, an external approach, involves dissection and removal of the tumor through a scleral flap, while endoresection utilizes vitreoretinal surgical techniques for internal tumor excision. Both surgical techniques for uveal melanoma removal are highly challenging, carry significant risks of complications, and should be performed by experienced vitreoretinal surgeons specializing in ocular oncology at specialized centers. They aim to preserve the eye and potentially maintain useful vision while saving patients' lives, a crucial consideration in ocular oncology. These surgical modalities have become increasingly relevant as the field progresses towards more conservative, function-preserving approaches in cancer treatment. The aim of this article is to conduct a comparative analysis of exoresection and endoresection in the context of uveal melanoma management. By critically evaluating these surgical modalities, this review seeks to provide clinicians with valuable insights to guide treatment decisions for patients with large uveal melanomas.

Keywords:

Complications, operative therapy, treatment outcome, uveal melanoma

Introduction

The management of uveal melanoma is a complex decision-making process that takes into account multiple factors. Clinicians must consider the tumor's characteristics, such as its size, location, and any associated complications such as retinal detachment, vitreous hemorrhage, or retinal invasion. Equally important are patient-specific factors, including age, overall health status, the condition of the unaffected eye, and the individual's personal treatment preferences.^[1-4]

This is an open access journal, and articles are distributed under the terms of the Creative Commons Attribution-NonCommercial-ShareAlike 4.0 License, which allows others to remix, tweak, and build upon the work non-commercially, as long as appropriate credit is given and the new creations are licensed under the identical terms.

For reprints contact: WKHLRPMedknow_reprints@wolterskluwer.com

Observational management, primarily indicated for indeterminate lesions, may be considered for select small melanomas (<3 mm apical height, <10 mm basal diameter) under ocular oncologist supervision, with treatment mandated on documented growth.^[5] For larger tumors that require treatment, therapeutic options range from less invasive techniques, such as radiotherapy (plaque brachytherapy, proton beam therapy), transpupillary thermotherapy (TTT), photodynamic therapy, and local resection (either externally or internally), to more radical procedures such as enucleation or orbital exenteration. In some cases, a combination of these approaches is employed to achieve the best possible outcome.^[6] The Collaborative

*Address for correspondence:

Dr. Min Kim,
Department of
Ophthalmology, Institute
of Vision Research,
Gangnam Severance
Hospital, Yonsei University
College of Medicine, 211
Eonjuro, Gangnam-gu,
Seoul 06273,
South Korea.
E-mail: minkim76@
gmail.com

Submission: 16-10-2024

Accepted: 08-12-2024

Published: 06-02-2025

How to cite this article: Lee SM, Kim M. Surgical treatment of uveal melanoma: Exoresection versus endoresection. Taiwan J Ophthalmol 2025;15:34-44.

Ocular Melanoma Study (COMS) has influenced treatment protocols, recommending plaque radiotherapy over enucleation for most medium-sized melanomas,^[7,8] while also noting that preenucleation irradiation offers no added benefit.^[9,10] Notably, the tumor's genetic profile has been demonstrated to be a more significant determinant of prognosis than the specific treatment modality selected.^[11-13]

In recent decades, the demand for eye-preserving treatments in uveal melanoma management has increased, with radiation therapy emerging as one of the most widely utilized conservative approaches.^[14] Among the radiation modalities, Ruthenium-106 brachytherapy is typically indicated for lesions with an apical height not exceeding 6 mm and a basal diameter of <20 mm. Iodine-125 and palladium-103 plaques offer greater flexibility, accommodating tumors of varying thicknesses.

All of these treatments have both advantages and disadvantages. The efficacy of plaque brachytherapy is limited in cases where tumors surpass 12 mm in apical height or 20 mm in basal diameter, necessitating consideration of alternative therapeutic modalities.^[15] Furthermore, the application of brachytherapy in large uveal melanomas is constrained by the development of vision-threatening complications such as radiation retinopathy, papillopathy, and "toxic tumor syndrome," which further complicates treatment outcomes.^[11,16] Moreover, it is generally contraindicated for tumors in close proximity to the optic disc to minimize radiation-induced damage to the fovea and optic nerve.^[11,14] To mitigate radiation-related complications and enhance the likelihood of preserving functional vision, surgical tumor removal can be considered.^[11] In selected patients with uveal melanoma, surgical resection can be performed as a primary procedure to treat uveal melanoma while minimizing potential complications associated with plaque brachytherapy or as a secondary treatment. In particular, surgical resection is an especially attractive option for treating large uveal melanomas which might require enucleation or may lead to serious vision threatening complications from excessive radiation from brachytherapy. Surgical removal of tumors may lead to an increased chance of preserving functional vision and maintaining the globe. Surgical resection of uveal melanoma can be categorized into two primary approaches: Exoresection, an external en bloc resection through a scleral trapdoor, usually performed for large tumors located anteriorly involving iris and ciliary body; and endoresection, an ab interno transretinal removal of tumor with a vitreous cutter, for tumors located posteriorly.^[17]

However, the surgical treatment of large uveal melanoma is quite challenging and only performed at

selected ocular oncology centers throughout the world by experienced surgeons, owing to its technical difficulties, the controversial nature of the procedure regarding potential increased risk of local recurrence and systemic metastasis by dissemination, and potential serious ocular and systemic complications, including death.

The principal objective of this article is to describe each surgical technique and to conduct a comparative analysis of the advantages, disadvantages, clinical outcomes, and prognostic implications associated with these two surgical modalities in the treatment of uveal melanoma. Of note, written informed consent was obtained from the patients for the use of their clinical data and images in the journal. All figures presented herein are original and have not been previously published.

Exoresection (Transscleral Resection, Ab externo)

Background

Local resection of uveal melanoma was first documented in 1911.^[18] Stallard later developed a technique in the 1960s involving scleral flap creation and diathermy, which he termed "choroidectomy."^[19] In the 1970s, with the advent of microsurgical techniques and instruments, Foulds developed a novel approach for tumor excision. This method involved performing scleral lamellar dissection at the tumor's base, removing the tumor with partially dissected interior scleral lamella, followed by closure of the external scleral lamella.^[20-25]

The COMS confirmed that brachytherapy did not increase the risk of metastasis and mortality rates compared to enucleation, thus supporting the use of eye-preservation strategies.^[7,8] However, due to the increased incidence of toxic tumor syndrome after irradiation of large tumors, surgical resection emerged as an alternative treatment modality.

Indications

Exoresection is usually indicated for patients who refuse to undergo enucleation, have poor vision in the contralateral eye and, and who are deemed likely to respond poorly to radiotherapy or likely to develop serious postradiation complications due to the tumor's large size. Exoresection is particularly suitable for tumors with a height >7 mm, a largest tumor diameter <22.0 mm, anterior location of the tumor, and a posterior tumor margin >3 mm from the fovea or optic disc. Furthermore, patients should be in good general health to tolerate hypotensive anesthesia.^[26-29] Exoresection is contraindicated for those who cannot tolerate hypotensive anesthesia due to systemic conditions (ischemic heart disease, cerebrovascular disease, retinal or respiratory impairment, or any other disease that can

be compromised by the hypotensive anesthesia during the surgery), basal tumor diameter >18 mm, extensive retinal invasion, and extraocular extension. Absolute contraindications include optic nerve invasion and diffuse melanoma.

Surgical technique

The most widely utilized exoresection technique is partial lamellar sclerouvectomy (PLSU). It begins with a 180° limbal conjunctival peritomy, followed by removal of the episcleral tissue. When needed, extraocular muscles are disinserted. It is usually helpful to leave approximately 1 mm tendon stump for reattachment of the muscle at the time of plaque removal. Traction sutures are then placed on the sclera and tumor margins are identified through transillumination and marked on the sclera. To reduce the risk of bleeding, vortex veins overlying the intended scleral flap are cauterized. A partial thickness scleral flap overlying the tumor is made. For good wound edge apposition, it is usually helpful to create this flap in a polyhedral shape rather than circular. The scleral flap is typically created at a distance of 5 mm from the tumor margin. However, for a very large tumor, this can result in an excessively large scleral flap, complicating an already challenging procedure. Based on my personal experience, for large choroidal melanomas with a basal diameter >16 mm, the clearance zone between the tumor and flap margin can be slightly adjusted according to the tumor size. Then, a posteriorly hinged lamellar scleral flap, approximately 80% of the scleral thickness, is created. Care must be taken to prevent the formation of buttonholes while creating lamellar scleral flap. Any inadvertent buttonholes during the procedure must be sutured immediately to ensure no tumor prolapse. Long ciliary vessels overlying the scleral flap as well as the short ciliary vessels near the optic nerve may be cauterized to minimize the risk of intraoperative hemorrhage. A deep scleral incision is then made to expose the choroid; buttonhole incisions are first created, approximately 2 mm inside the superficial lamellar scleral flap anterolateral to the tumor. This stepped wound edge would help achieve tight closure of the wound when suturing. The scleral incision is extended with scissors with the anterior margin being incised last. This prevents prolapse of the intraocular contents during incision. Before proceeding with the tumor excision, one must mark the deep scleral lamella so that the orientation of the removed tumor can be easily identified later for a pathological analysis. Cauterizing the choroid around the tumor with bipolar cautery or laser may help reduce intraoperative hemorrhage. The tumor is carefully removed by holding the deep scleral lamellar flap with forceps with one hand and gently separating them apart with the other hand with forceps or cotton-tip applicator to divide strands of tissue joining the tumor to the retina, allowing the retina to naturally

detach from the tumor. Blunt-tipped scissors can also be used to separate the tumors by blunt dissection. The tumor is removed along with the deep lamellar scleral flap, which remains adherent to the tumor. Care must be taken to remove the choroidal tumor without disrupting the underlying retina or vitreous to minimize postoperative complications. Retinal breaks typically develop during the stage when the tumor is separated from the adherent retina.^[6,29]

Regardless of all the measures taken to prevent hemorrhage during the procedure, there is usually some bleeding, and it is most helpful to achieve hemostasis and remove any clots at this time. After the tumor is removed, the instrument set is changed for a new one to prevent potential tumor seeding. The scleral flap is then repositioned and sutured with nonabsorbable 8-0 nylon interrupted sutures. Hypotensive anesthesia, maintaining systolic blood pressure around 40–50 mmHg, is used throughout the procedure to minimize intraoperative bleeding. If there is any transscleral extension of tumor, it can be excised together with superficial lamellar scleral flap and any residual tumor can be treated with adjunctive plaque brachytherapy. Plaque brachytherapy, if available, is usually administered as part of the standard protocol, delivering a dose of 85–95 Gy to a depth of 2 mm. At our center, ruthenium-106 plaque is used for this purpose. Adjunctive plaque brachytherapy plays a critical role in reducing complications associated with wide surgical resection margins, as it ensures radiotherapy fully encompasses the surgically resected areas, thereby minimizing the risk of recurrence and wound dehiscence. In some cases, vitrectomy may be performed to lower intraocular pressure, although this is not always necessary.^[6,29] Postoperatively, the patient is positioned to prevent accumulation of any subretinal hemorrhage under the fovea.

Outcomes

Efficacy in preserving vision and eye-globe

Several studies comparing exoresection to iodine brachytherapy found that exoresection resulted in better visual outcomes and greater vision retention.^[26,30-32] Foulds and Damato studied 134 patients with large choroidal melanomas (up to 15 mm in diameter) who underwent exoresection, and found the procedure achieved favorable globe preserving outcomes in 82% of cases, with 60% of patients retaining useful vision and 25% attaining good vision (6/6–6/18).^[33] In another study by Foulds *et al.*, 157 patients treated with exoresection demonstrated that 65% retained useful vision, ranging from 6/6 to counting fingers.^[21] Damato *et al.* identified key preoperative factors predicting the retention of good vision (6/12 or better), including nasal tumor location and a distance of more than 1 disc diameter (DD) between the tumor and the optic disc or fovea. In contrast,

posterior tumor extension within 1 DD of the optic disc and/or fovea was the most significant predictor of severe visual loss.^[34]

Limited head-to-head comparisons exist between exoresection and brachytherapy for choroidal melanoma. Two notable studies by Bechrakis *et al.*^[26] and Kivelä *et al.*^[30] demonstrated superior visual outcomes with transscleral resection compared to iodine-125 brachytherapy, while maintaining comparable disease-free survival and enucleation rates.

In a matched cohort study, Bechrakis *et al.*^[26] compared 18 patients treated with iodine brachytherapy to 36 patients who underwent transscleral resection. Their findings revealed that visual acuity of 20/200 or better was preserved in 61.1% of transscleral resection patients compared to only 5.6% of patients who received iodine brachytherapy.

Kivelä *et al.*^[30] conducted a comparative analysis between transscleral resection and iodine brachytherapy in patients with choroidal melanoma ≥ 6 mm in thickness. While the risk of vision loss to 20/60 showed no statistically significant difference between the two procedures, the risk of deterioration to 20/200 vision was significantly higher in the iodine brachytherapy group compared to the transscleral resection group. Surgical resection combined with plaque brachytherapy may reduce the incidence of radiation retinopathy and toxic tumor syndrome, particularly in large tumors. Treating such significantly large tumors with plaque brachytherapy alone would require excessive amount of radiation, which would inevitably lead to the development of radiation retinopathy and subsequent vision loss. Surgical removal of large tumors combined with adjunctive plaque brachytherapy can result in better visual outcomes, while disease-free survival and enucleation rates remain comparable to those achieved with plaque brachytherapy alone.

Surgery-related complications including local tumor control

The most common intraoperative complications include intraocular hemorrhage including vitreous hemorrhage, hyphema, and retinal or subretinal hemorrhage. In addition, choroidal detachment and retinal detachment are frequently observed adverse events.^[24] Inadvertent scleral perforation may occur during lamellar scleral dissection, and scleral flap loss could occur while separating the tumor from the retina with the scleral flap. If meticulous cauterization is not performed, uncontrolled bleeding may occur. Postoperative complications following exoresection include vitreous hemorrhage, retinal or subretinal hemorrhage, retinal detachment, cataract, and choroidal

detachment [Figure 1]. Among these, cataract and retinal detachment typically necessitate additional surgical interventions.^[24,35] If vitreous hemorrhage persists or retinal detachment occurs, timely vitreoretinal surgery is essential to prevent serious complications such as proliferative vitreoretinopathy (PVR). Damato *et al.* studied 156 patients and found rhegmatogenous retinal detachment in 28 (18%) eyes. Among the 25 patients who had surgery, 21 (84%) achieved successful anatomic outcomes.^[35] Scleral thinning or wound dehiscence, often consequent to suboptimal flap dissection, presents a notable postoperative concern. While typically manageable through observation, pronounced uveal tract bulging may necessitate scleral reinforcement through patch grafting.^[36,37]

Postoperatively, tumor recurrence is one of the serious complications. Tumor recurrence at the primary site may result from two main factors: Insufficient excision of neoplastic tissue or iatrogenic dissemination of tumor cells during intraoperative manipulation of the globe.^[29] Bechrakis *et al.* reported that the 5-year and 10-year local recurrence rates after exoresection with adjuvant brachytherapy were 24% and 32%, respectively.^[27] Puusaari *et al.* reported 5-year local recurrence rates of 37% for exoresection with adjuvant therapy versus 7% for iodine-125 plaque brachytherapy.^[31] Kivelä *et al.* conducted a case-control study and observed local recurrence in 33% of the exoresection group and 6% of the brachytherapy-only group. However, the recurrence rate decreased to 13% in patients who underwent exoresection with adjuvant radiotherapy.^[30] Damato *et al.* reported 4-year recurrence rates ranging from 6% to 57%, contingent on identified risk factors. These factors included epithelioid cellularity, posterior tumor extension, large tumor diameter, and absence of ruthenium plaque radiotherapy.^[38] Due to its ability to lower recurrence rates, adjuvant radiotherapy is recommended after tumor removal; however, adjuvant brachytherapy may cause wound dehiscence, optic neuropathy, and radiation maculopathy. To reduce these risks, resected margins should be closed with nonabsorbable sutures, and the plaque should be positioned at an adequate distance from both the optic nerve and macula to minimize radiation-induced damage to these critical structures.^[29] Other postoperative complications include hyphema, increased intraocular pressure, strabismus, corneal edema, cataract, and endophthalmitis. Late postoperative complications include development of PVR, bullous keratopathy, and phthisis.

Metastasis and survival rate

Metastasis rates following exoresection for posterior uveal melanomas varied across studies, with Bechrakis *et al.* reporting 28% and 44% at 5 and 10 years'

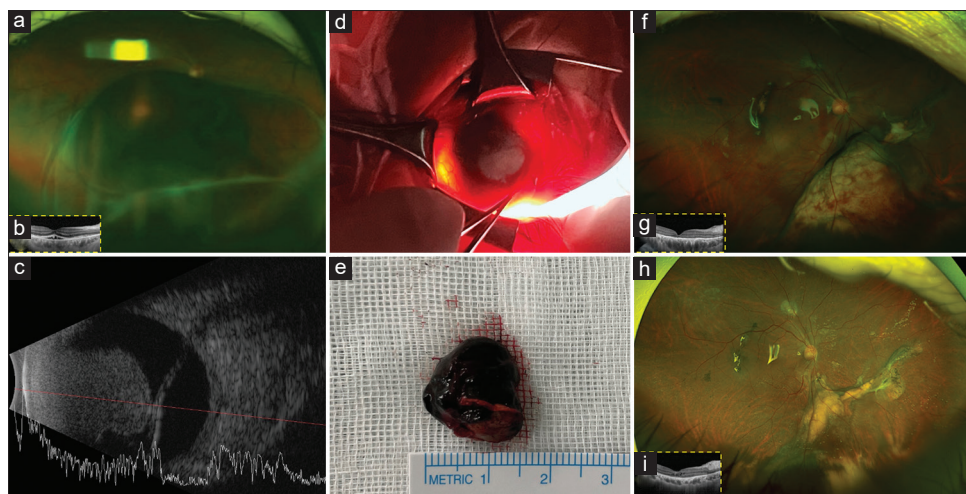


Figure 1: Depicts the case of a 59-year-old male with choroidal melanoma in the right eye. Preoperative imaging reveals a large, mushroom-shaped choroidal tumor located inferiorly, associated with bullous retinal detachment (a-c). The tumor measured 18 mm in maximal diameter, 11 mm in minimal diameter, and 13 mm in height, with macular involvement and an initial visual acuity of 20/100. The patient underwent exoresection, with scleral transillumination showing the tumor shadow (d) and the excised tumor specimen, (e) Three months post-exoresection, fundus examination demonstrated a flat retina with large inferonasal subretinal haemorrhage, (f and g) At 1-year follow-up, significant absorption of the subretinal hemorrhage was observed, with maintained retinal attachment (h and i). No local tumor recurrence or distant metastasis was detected, though final visual acuity was <20/200

posttreatment respectively,^[27] while Shields *et al.* observed a 5.3% incidence over a 2.5-year median follow-up.^[24]

Studies have reported no significant differences in survival rates between patients undergoing exoresection and those treated with enucleation or brachytherapy.^[21,26,30,32,39] Prescher *et al.* found that chromosome 3 loss, tumor size ≥ 15 mm, and ciliary body involvement were indicators of increased risk for metastasis and death.^[40] Damato *et al.* added age >60 years, presence of epithelioid cells, tumor diameter >16 mm, superior localization, and absence of adjuvant brachytherapy as significant predictors of metastatic death.^[38] These factors are linked to tumor characteristics rather than treatment methods. Consequently, exoresection could be considered a viable option based on individual patient and tumor factors, especially when treating a large uveal melanoma, untreatable with conventional treatment modality except enucleation.

Endoresection (Ab interno)

Background

Endoresection is an internal approach for the removal of uveal melanoma. It involves a transretinal tumor resection or peripheral retinectomy followed by the piecemeal removal of the tumor. Using a vitreous cutter, the tumor is fragmented into small pieces and removed. This procedure is technically challenging, requiring advanced vitreoretinal surgical techniques, and is therefore performed only in specialized centers with expertise in both ocular oncology and vitreoretinal surgery.

Peyman and Cohen first introduced an ab interno approach for the management of uveal melanoma located within 1–2 DD from the optic disc, describing it as retinochoroidectomy.^[41] Subsequently, Damato *et al.* adopted and termed this technique “endoresection.”^[42] Over the past three decades, advancements in vitreoretinal surgery have facilitated the application of endoresection.^[43]

Indications

Endoresection has gained acceptance as a management strategy for uveal melanomas that are unsuitable for brachytherapy, particularly those adjacent to the optic nerve or macula, as well as large posterior melanomas in patients who refuse enucleation. Although traditionally contraindicated for ciliary body involvement, the endoresection technique can be effectively applied to melanomas located anteriorly such as ciliochoroidal melanoma, in select cases based on the author’s personal experience. Endoresection may also be performed as a primary salvage therapy following plaque brachytherapy or concurrently with radiotherapy. However, it is generally recommended to perform radiotherapy first before initiating endoresection due to the potential risk of disseminating malignant tumor cells. This surgical approach is employed to minimize radiotherapy-related morbidity and maintain visual function, especially in monocular patients.^[6,29,44]

Surgical technique

As with any surgical procedure, the importance of thorough preoperative preparation in endoresection cannot be overstated. If hypotensive anesthesia is planned, consultation with an anesthesiologist is essential,

particularly for patients with ischemic cardiovascular conditions. Ensuring comfortable positions for both the patient and surgeon is critical as endoresection tends to take longer than standard vitreoretinal surgery to minimize surgical stress. The surgeon must be fully acquainted with the tumor's configuration and dimensions through preoperative fundus examination, transillumination, and magnetic resonance imaging. Based on this information and the tumor's quadrant location, the surgeon's position should be adjusted at the start of the surgery. Continuous communication with the anesthesiologist is vital throughout the procedure to monitor the patient's vital signs, as serious complications such as venous air embolism have been reported during endoresection.

Similar to exoresection, it is highly recommended to perform transillumination before making any intraocular access to delineate the tumor margin, particularly when adjunctive plaque brachytherapy is planned. Muscles are isolated and disinserted where necessary, and cauterization of the vortex vein in the quadrant of the tumor is crucial to prevent potential air embolism. These steps are important for minimizing intraoperative complications and ensuring optimal surgical outcome. Then, the endoresection procedure typically begins with a lensectomy or phacoemulsification, followed by intraocular lens implantation. I recommend performing a combined phacovitrectomy for removal of cataract during the initial surgery, as it allows better access to the peripheral vitreous and tumors located anteriorly. A conventional 20–23-gauge vitreous cutter under wide-field visualization is used to perform a thorough vitrectomy. While a 25-gauge cutter may suffice for removal of smaller tumors, I recommend using a 23-gauge cutter for endoresection. The larger port size facilitates more efficient tumor removal and provides better control over the intraocular environment, particularly for managing bleeding during the procedure. A fourth port for supplementary endoillumination, utilizing a chandelier system, is advisable to enable bimanual techniques during the endoresection procedure. Complete posterior vitreous detachment is achieved, ensuring that the posterior hyaloid is detached from the anterior tumor margin. Before proceeding with the surgical resection of the tumor, it is crucial to meticulously remove the posterior hyaloid and shave the peripheral vitreous to prevent postoperative PVR formation. Performing peripheral shaving after tumor resection can be extremely challenging; severe intraoperative bleeding and development of retinal detachment can hinder the surgeon's ability to achieve complete shaving of the peripheral vitreous and posterior hyaloid.

Endodiathermy of retinal vessels over the tumor base is performed, and endolaser cauterization of the tumor

margin may be conducted, unless it is inaccessible due to the tumor's large size. Preoperative endolaser retinopexy around the tumor margin would be most useful in preventing progression of retinal detachment, but it is usually difficult to perform when the tumor is too large and accompanied by extensive serous retinal detachment. 1–2 cc of perfluorocarbon liquid (PFCL) is injected over the posterior pole to prevent the migration of subretinal hemorrhage and the progression of retinal detachment during surgery. The tumor can be removed in two ways: It can be removed directly transretinally along with any invaded retina, or a peripheral retinectomy can be performed first, allowing the retina to be folded away from the tumor if there is no retinal invasion of the tumor. The tumor can then be removed in a piecemeal fashion using the vitreous cutter. I recommend using a low cutting rate (500–1000 cuts/min) and a high aspiration rate (~650 mmHg) for efficient tumor fragmentation and aspiration. Creating a deep choroidal tunnel with the vitreous cutter first, followed by piecemeal removal of the tumor, helps control intraoperative hemorrhage. This technique allows any bleeding to accumulate within the choroidal tunnel, which can be readily aspirated, rather than dispersing through the vitreous cavity, and obstructing the surgeon's view of the fundus. Tumor excision is performed under elevated intraocular pressure (up to 85 mmHg) and controlled systemic hypotension (systolic blood pressure ≤ 90 mmHg) to minimize hemorrhage from tumor vasculature. Controlling intraoperative bleeding is crucial to the success of the surgery.^[6,29,37,44]

Additional PFCL is injected to stabilize and flatten the retina, followed by endolaser treatment of the retina, choroidal resection edges, and exposed scleral surface to destroy any residual tumor cells. Scleral bed and edges are cleaned with the vitreous cutter and endolaser to destroy any residual tumor. This step is crucial for minimizing hemorrhage and eradicating potential tumor remnants, especially if adjunctive plaque brachytherapy is not available. To prevent retinal re-detachment, postoperative hemorrhage, and the risk of venous air embolism, a direct PFCL-silicone oil exchange is performed. The procedure concludes with cryotherapy applied to the sclerotomy sites and adjuvant plaque brachytherapy in selected cases.^[6,29,37,44] Personally, I advocate for adjunctive plaque brachytherapy in all cases regardless of the tumor's clinical features. It is extremely difficult to identify any residual tumor cells visually. Performing adjunctive plaque brachytherapy ensures complete treatment of any remaining tumor cells.

Outcomes

Efficacy in preserving vision and eye-globe

Endoresection has demonstrated efficacy in preserving functional vision for patients with uveal melanoma. In a

study by Kertes *et al.*, 31.2% of patients maintained visual acuities of 6/60 or better, 56.3% retained vision between 6/120 and light perception, while 12.5% experienced complete loss of light perception.^[45] Konstantinidis *et al.* reported that 13% of cases achieved visual acuity better than 6/12, and 31% better than 6/30.^[46] Another study by Biewald *et al.* revealed that 13.4% of patients retained vision of 20/50 or better, 33.6% had visual acuity ranging from 20/400 to 20/50, and 53% exhibited visual acuity worse than 20/400.^[11] Rice *et al.* conducted a comparative study between brachytherapy and endoresection in the treatment of medium-sized choroidal melanoma. Their findings indicated that patients in the endoresection group demonstrated a higher probability of achieving better final visual acuity.^[47] Studies have reported an eye preservation rate of over 90% following endoresection for uveal melanoma.^[11,42,48]

Surgery-related complications including local tumor control

Surgical interventions for uveal melanoma are associated with a range of potential complications. The most frequently observed adverse events include scleral bed hemorrhage, cataract formation, ocular hypertension, retinal detachment, vitreous hemorrhage, macular traction syndrome, epiretinal membrane development, band keratopathy, and phthisis bulbi. Secondary complications, often attributed to adjuvant radiotherapy, encompass retinopathy and optic nerve atrophy. Postoperative management frequently involves additional surgical interventions, with cataract extraction and silicone oil removal being the most commonly performed procedures.^[11,42,45,46,48-53] Subfoveal PFCL droplet retention, while uncommon, can occur following endoresection due to the extensive retinal manipulation involved, which allows small PFCL droplets to migrate beneath the large resected retinal flap during surgery [Figure 2].

Vascular air embolism is defined as the entry of air or gas from the operative field into the venous or

arterial circulation, resulting in systemic sequelae. The endoresection technique for uveal melanoma, which involves excision of choroidal tissue in the tumor region, potentially exposes open choroidal and tumor vasculature to infused air. A recent report by Ruschen *et al.* documented a fatal case of gas embolism associated with perfluorooctane (PFO) use, despite the absence of intraoperative air or gas introduction. The authors hypothesized that this embolism resulted from PFO entering the bloodstream and subsequently forming gas bubbles in the pulmonary circulation, possibly due to PFO's elevated vapor pressure. To mitigate these risks, the avoidance of air infusion and the ligation of vortex veins draining the excision site are recommended.^[54-62]

Studies report a wide range of local recurrence rates following endoresection for uveal melanoma, with incidences varying from 0% to 22.9%.^[11,46,49,53] Comparative studies have shown similar local recurrence rates between endoresection and brachytherapy.^[47,63] Notably, late recurrences can occur, as evidenced by Mittica *et al.*'s report of local recurrence 13 years postinternal resection.^[64] Adjuvant brachytherapy postendoresection is associated with reduced local recurrence rates, which correlates with lower metastasis rates. Consequently, adjuvant brachytherapy is frequently employed in postendoresection management of uveal melanoma.^[38,63]

In either surgical treatment modality, complete local tumor control may not always be achieved and recurrence can occur either at the margin of the surgically treated area, within the treated zone or rarely, as a satellite lesion distant from the original tumor. Early detection of recurrent or residual tumors can be challenging, as they must be differentiated from reactive RPE hyperplasia, pigment laden macrophages, and organized subretinal hemorrhages. Serial multimodal imaging including optical coherence tomography is crucial in identifying these lesions and distinguishing them from true recurrence of the tumor. Failure to correctly identify recurrence could result in extraocular

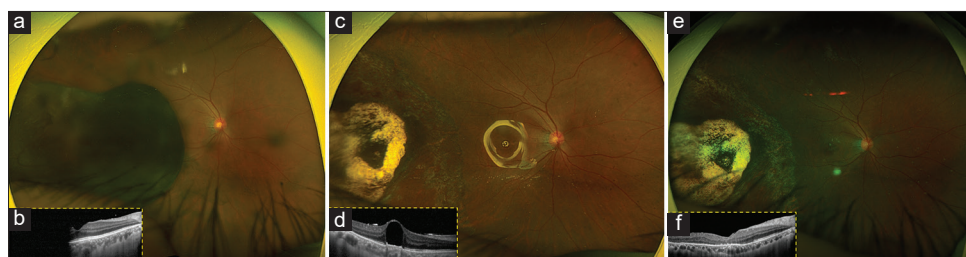


Figure 2: Illustrates the case of a 51-year-old male with choroidal melanoma in the right eye. Initial examination revealed a mushroom-shaped, pigmented choroidal mass on the temporal side of the fundus, measuring 16 mm in maximal diameter, 12 mm in minimal diameter, and 11 mm in height, with associated overlying hemorrhage and exudative retinal detachment. (a) Optical coherence tomography demonstrated a flat fovea, though the temporal aspect was obscured by the tumor's shadow, (b) Preoperative visual acuity was 20/20. The patient underwent endoresection, after which no local tumor recurrence or distant metastasis was observed. However, a subfoveal perfluorocarbon liquid (PFCL) droplet was detected postoperatively, (c and d) A second vitrectomy was performed to remove the PFCL droplet, involving induced retinal detachment with a 41-gauge fine needle and balanced salt solution, droplet aspiration, and 20% sulfur hexafluoride gas tamponade. At 6-month follow-up postsecond surgery (1 year postendoresection), there was no evidence of local recurrence or metastasis. The retina remained flat, but outer retinal disruption at the fovea resulted in vision worse than 20/200 (e and f)

extension and potentially necessitate enucleation, if not detected in time.

Local recurrence at the surgical resection margin can be treated with TTT. For extensive local recurrences not amenable to TTT, treatment options include repeat plaque brachytherapy or additional surgical resection performed either internally or externally, followed by adjunctive plaque brachytherapy. Enucleation is typically reserved as a last resort for extensive or diffuse tumor recurrence that are not readily treatable with the aforementioned treatment modalities.

Metastasis and survival rate

Reported mortality rates due to metastasis following endoresection for uveal melanoma vary considerably, ranging from 0% to 20%.^[11,42,45,49] Konstantinidis *et al.* observed all-cause mortality rates of 9% at 5 years and 20% at 10 years' posttreatment.^[46] Two comparative studies between primary endoresection and primary brachytherapy have revealed no significant differences in overall survival or relapse-free survival.^[47,63] These findings suggest that endoresection may offer a viable alternative to brachytherapy in selected cases, potentially preserving visual function without compromising oncological outcomes.

We conducted a retrospective review of 31 endoresections performed at our center over the past 5 years (unpublished data). The patient cohort had a mean age at diagnosis of 59.8 years (range: 29–81 years). The tumors presented with a mean largest diameter of 14.4 mm (range: 8.4–25.7 mm) and a mean thickness of 9.0 mm (range: 4.5–14.7 mm). Several challenging cases were encountered, including a patient with a solitary functioning eye who declined enucleation, and a case involving a large choroidal melanoma exceeding 20 mm in diameter and 10 mm in thickness, accompanied by total retinal detachment and advanced glaucoma.

The globe salvage rate following endoresection was 96% (30 out of 31). Postoperatively, 45.2% of patients (14 out of 31) achieved a mean visual acuity of 20/200 or better. Local tumor recurrence was observed in two eyes (6.5%) after a mean follow-up period of 3 months, and 10 months, respectively. Metastatic disease developed in four cases: Three instances of liver metastasis within 2 years postoperatively, and one case of conjunctival metastasis at 1 year postoperatively.

Discussion

The surgical treatment of large uveal melanomas is quite challenging, requiring expertise in both ocular oncology and vitreoretinal surgery. Recent advancements in vitreoretina and oncology have enabled better eye

preservation and local tumor control for large uveal melanoma. Specifically, exoresection and endoresection for the treatment of large uveal melanomas offer the potential for eye and vision preservation. Exoresection is particularly suitable for anteriorly located and larger tumors, whereas endoresection presents advantages for tumors in close proximity to the optic disc or macula. The decision to perform surgical removal of uveal melanomas is complex and typically reserved for tumors where other treatments, such as radiation therapy, are less effective or pose higher risks. It is generally carried out in specialized centers with expertise in both ocular oncology and vitreoretinal surgery.

In endoresection of uveal melanomas, concerns have been raised about the potential for both systemic and local dissemination of malignant cells due to the piecemeal removal of the tumor.^[65] During this procedure, the tumor's integrity is compromised as it is fragmented into smaller pieces, unlike in PLSU, where the tumor is removed en bloc. This fragmentation contradicts the fundamental surgical principle of complete, intact tumor removal in modern cancer surgery, contributing to the ongoing controversy surrounding endoresection. However, genetic studies suggest that metastasis is primarily a biological process, occurring independently of surgical intervention.^[40,66,67] Systemic metastasis is more closely linked to tumor characteristics than to treatment methods. To date, there is no concrete evidence proving that the surgical intervention influences metastasis. It appears that the fate of the tumor and the patient's survival are already determined before surgical resection takes place. With advancements in molecular prognostication, clinical features of uveal melanoma – such as ciliary body involvement, large tumor diameter and thickness, retinal invasion, histopathological finding of predominant epithelioid cell type, and tumor cytogenetics – play a key role in determining the prognosis. The risk of melanoma seeding at sclerotomy sites remains, which necessitates precautionary measures such as the application of external cryotherapy to these sites.^[44,68]

Regarding local tumor control, recent studies suggest that endoresection, when combined with adjunctive treatments like adjuvant brachytherapy, can provide good local control in selected cases of uveal melanoma without necessarily increasing the risk of metastasis. Adjuvant radiotherapy, such as Ruthenium-106 brachytherapy, is often used to reduce the risk of microscopic tumor seeding. However, concerns remain about whether the physical manipulation of the tumor during endoresection could contribute to metastasis, particularly in cases where radiotherapy is not applied comprehensively. Furthermore, it can be extremely

challenging to assess the resected tumor margins and accurately differentiate normal choroidal tissue from choroidal tumor during surgery, making it difficult to ensure complete tumor removal. As a result, there is a risk of incomplete tumor resection and subsequent tumor recurrence after the surgery.

Besides surgical treatments, neoadjuvant therapy has also been introduced as a method for preserving the eye globe in cases of large uveal melanoma. Hiong *et al.* reported a case where neoadjuvant chemotherapy was successfully utilized in a patient with a large uveal melanoma that initially surpassed the size criteria for primary brachytherapy. The patient, with pre-existing unilateral blindness in the left eye, required a conservative approach to preserve vision in the right eye affected by uveal melanoma. The tumor dimensions at presentation were 18 mm in maximal diameter and 16.5 mm in maximal thickness. Therefore, the medical team administered neoadjuvant darovasertib (protein kinase C inhibitor) and crizotinib (mitogen-activated protein kinase/extracellular signal-regulated kinase inhibitor), a regimen recognized for its efficacy in controlling metastatic uveal melanoma. This intervention effectively reduced the tumor dimensions to 14.1 mm and 2.6 mm. This substantial reduction in tumor size facilitated the subsequent application of plaque brachytherapy, thereby avoiding the need for enucleation.^[69]

The efficacy of this neoadjuvant approach in primary uveal melanoma treatment, if further validated, could potentially establish itself as an additional globe-conserving modality alongside surgical resection for large uveal melanomas. This development represents a promising avenue for expanding the therapeutic armamentarium in managing challenging cases of uveal melanoma.

Conclusion

With the advancement of surgical techniques and equipment, surgical resection has emerged as a promising approach for managing large uveal melanomas that exceed traditional size criteria for plaque brachytherapy or pose a high risk of radiation-related complications. These techniques offer enhanced potential for vision preservation and globe retention while providing effective tumor control, especially when combined with adjunctive radiation therapy. Ongoing research in this field continues to broaden the therapeutic landscape, potentially improving patient outcomes and quality of life.

Data availability statement

Data sharing not applicable to this article as no datasets were generated or analyzed during the current study.

Financial support and sponsorship

Nil.

Conflicts of interest

The authors declare that there are no conflicts of interests of this paper.

References

1. Damato B. Progress in the management of patients with uveal melanoma. The 2012 Ashton Lecture. Eye (Lond) 2012;26:1157-72.
2. Damato B, Heimann H. Personalized treatment of uveal melanoma. Eye (Lond) 2013;27:172-9.
3. Shields JA, Shields CL. Management of posterior uveal melanoma: Past, present, and future: The 2014 Charles L. Schepens lecture. Ophthalmology 2015;122:414-28.
4. Dogrusöz M, Jager MJ, Damato B. Uveal melanoma treatment and prognostication. Asia Pac J Ophthalmol (Phila) 2017;6:186-96.
5. Weis E, Surgeoner B, Salopek TG, Cheng T, Hyrcza M, Kostaras X, *et al.* Management of uveal melanoma: Updated cancer care Alberta clinical practice guideline. Curr Oncol 2023;31:24-41.
6. Sadda SR, Schachat AP, Wilkinson CP, Hinton DR, Wiedemann P, Freund KB, *et al.* Ryan's Retina. Philadelphia: Elsevier Health Sciences; 2022.
7. Diener-West M, Earle JD, Fine SL, Hawkins BS, Moy CS, Reynolds SM, *et al.* The COMS randomized trial of iodine 125 brachytherapy for choroidal melanoma, III: Initial mortality findings. COMS Report No. 18. Arch Ophthalmol 2001;119:969-82.
8. Collaborative Ocular Melanoma Study Group. The COMS randomized trial of iodine 125 brachytherapy for choroidal melanoma: V. Twelve-year mortality rates and prognostic factors: COMS report No. 28. Arch Ophthalmol 2006;124:1684-93.
9. The Collaborative Ocular Melanoma Study (COMS) randomized trial of pre-enucleation radiation of large choroidal melanoma II: Initial mortality findings. COMS report no. 10. Am J Ophthalmol 1998;125:779-96.
10. Hawkins BS, Collaborative Ocular Melanoma Study Group. The Collaborative Ocular Melanoma Study (COMS) randomized trial of pre-enucleation radiation of large choroidal melanoma: IV. Ten-year mortality findings and prognostic factors. COMS report number 24. Am J Ophthalmol 2004;138:936-51.
11. Biewald E, Lautner H, Gök M, Horstmann GA, Sauerwein W, Flühs D, *et al.* Endoresection of large uveal melanomas: Clinical results in a consecutive series of 200 cases. Br J Ophthalmol 2017;101:204-8.
12. Honavar SG. Is Collaborative Ocular Melanoma Study (COMS) still relevant? Indian J Ophthalmol 2018;66:1385-7.
13. Damato BE. Treatment selection for uveal melanoma. Dev Ophthalmol 2012;49:16-26.
14. Brewington BY, Shao YF, Davidorf FH, Cebulla CM. Brachytherapy for patients with uveal melanoma: Historical perspectives and future treatment directions. Clin Ophthalmol 2018;12:925-34.
15. American Brachytherapy Society – Ophthalmic Oncology Task Force Electronic address: paulfingher@eyecancercom, ABS – OOTF Committee. The American Brachytherapy Society consensus guidelines for plaque brachytherapy of uveal melanoma and retinoblastoma. Brachytherapy 2014;13:1-14.
16. Damato B. Developments in the management of uveal melanoma. Clin Exp Ophthalmol 2004;32:639-47.
17. Fallico M, Raciti G, Longo A, Reibaldi M, Bonfiglio V, Russo A, *et al.* Current molecular and clinical insights into uveal melanoma (Review). Int J Oncol 2021;58:10.
18. Zirm E. About endobulbar operations. Arch Augenheilkd 1911;69:233-51.
19. Stallard HB. Partial choroidectomy. Br J Ophthalmol 1966;50:660-2.

20. Foulds WS. The local excision of choroidal melanomata. *Trans Ophthalmol Soc U K* (1962) 1973;93:343-6.
21. Foulds WS, Damato BE, Burton RL. Local resection versus enucleation in the management of choroidal melanoma. *Eye (Lond)* 1987;1:676-9.
22. Shields JA, Augsburger JJ, Stefanyshyn MA, Connor RW. Sclerochorioretinal resection for choroidal melanoma. A clinicopathologic correlation of a postmortem eye. *Ophthalmology* 1984;91:1726-30.
23. Shields JA, Shields CL. Surgical approach to lamellar sclerouvectomy for posterior uveal melanomas: The 1986 Schoenberg lecture. *Ophthalmic Surg* 1988;19:774-80.
24. Shields JA, Shields CL, Shah P, Sivalingam V. Partial lamellar sclerouvectomy for ciliary body and choroidal tumors. *Ophthalmology* 1991;98:971-83.
25. Damato B. The role of eyewall resection in uveal melanoma management. *Int Ophthalmol Clin* 2006;46:81-93.
26. Bechrakis NE, Bornfeld N, Zöller I, Foerster MH. Iodine 125 plaque brachytherapy versus transscleral tumor resection in the treatment of large uveal melanomas. *Ophthalmology* 2002;109:1855-61.
27. Bechrakis NE, Petousis V, Willerding G, Krause L, Wachtlin J, Stroux A, *et al.* Ten-year results of transscleral resection of large uveal melanomas: Local tumour control and metastatic rate. *Br J Ophthalmol* 2010;94:460-6.
28. Romano MR, Catania F, Confalonieri F, Zollet P, Allegrini D, Sergenti J, *et al.* Vitreoretinal surgery in the prevention and treatment of toxic tumour syndrome in uveal melanoma: A systematic review. *Int J Mol Sci* 2021;22:10066.
29. Caminal JM, Lorenzo D, Gutierrez C, Slocker A, Piulats JM, Cobos E, *et al.* Local resection in choroidal melanoma: A review. *J Clin Med* 2022;11:7156.
30. Kivelä T, Puusaari I, Damato B. Transscleral resection versus iodine brachytherapy for choroidal malignant melanomas 6 millimeters or more in thickness: A matched case-control study. *Ophthalmology* 2003;110:2235-44.
31. Puusaari I, Damato B, Kivelä T. Transscleral local resection versus iodine brachytherapy for uveal melanomas that are large because of tumour height. *Graefes Arch Clin Exp Ophthalmol* 2007;45:522-33.
32. Caminal JM, Padrón-Pérez N, Arias L, Masuet-Aumatell C, Gutiérrez C, Piulats JM, *et al.* Transscleral resection without hypotensive anaesthesia versus iodine-125 plaque brachytherapy in the treatment of choroidal melanoma. *Eye (Lond)* 2016;30:833-42.
33. Foulds WS, Damato BE. Alternatives to enucleation in the management of choroidal melanoma. *Aust N Z J Ophthalmol* 1986;14:19-27.
34. Damato BE, Paul J, Foulds WS. Predictive factors of visual outcome after local resection of choroidal melanoma. *Br J Ophthalmol* 1993;77:616-23.
35. Damato B, Groenewald CP, McGalliard JN, Wong D. Rhegmatogenous retinal detachment after transscleral local resection of choroidal melanoma. *Ophthalmology* 2002;109:2137-43.
36. Kurt RA, Gündüz K. Exoresection via partial lamellar sclerouvectomy approach for uveal tumors: A successful performance by a novice surgeon. *Clin Ophthalmol* 2010;4:59-65.
37. Gündüz K, Bechrakis NE. Exoresection and endoresection for uveal melanoma. *Middle East Afr J Ophthalmol* 2010;17:210-6.
38. Damato BE, Paul J, Foulds WS. Risk factors for metastatic uveal melanoma after trans-scleral local resection. *Br J Ophthalmol* 1996;80:109-16.
39. Augsburger JJ, Lauritzen K, Gamel JW, DeBrakeleer DJ, Lowry JC, Eisenman R. Matched group study of surgical resection versus cobalt-60 plaque radiotherapy for primary choroidal or ciliary body melanoma. *Ophthalmic Surg* 1990;21:682-8.
40. Prescher G, Bornfeld N, Hirche H, Horsthemke B, Jöckel KH, Becher R. Prognostic implications of monosomy 3 in uveal melanoma. *Lancet* 1996;347:1222-5.
41. Peyman GA, Cohen SB. Ab interno resection of uveal melanoma. *Int Ophthalmol* 1986;9:29-36.
42. Damato B, Groenewald C, McGalliard J, Wong D. Endoresection of choroidal melanoma. *Br J Ophthalmol* 1998;82:213-8.
43. Reichstein D, Karan K. Endoresection utilizing pars plana vitrectomy for benign and malignant intraocular tumors. *Curr Opin Ophthalmol* 2019;30:151-8.
44. Damato BE. Local resection of uveal melanoma. *Dev Ophthalmol* 2012;49:66-80.
45. Kertes PJ, Johnson JC, Peyman GA. Internal resection of posterior uveal melanomas. *Br J Ophthalmol* 1998;82:1147-53.
46. Konstantinidis L, Groenewald C, Coupland SE, Damato B. Long-term outcome of primary endoresection of choroidal melanoma. *Br J Ophthalmol* 2014;98:82-5.
47. Rice JC, Stannard C, Cook C, Lecuona K, Myer L, Scholtz RP. Brachytherapy and endoresection for choroidal melanoma: A cohort study. *Br J Ophthalmol* 2014;98:86-91.
48. Vidoris AA, Maia A, Lowen M, Morales M, Isenberg J, Fernandes BF, *et al.* Outcomes of primary endoresection for choroidal melanoma. *Int J Retina Vitreous* 2017;3:42.
49. García-Arumí J, Sararols L, Martínez V, Corcostegui B. Vitreoretinal surgery and endoresection in high posterior choroidal melanomas. *Retina* 2001;21:445-52.
50. García-Arumí J, Zapata MA, Balaguer O, Fonollosa A, Boixadera A, Martínez-Castillo V. Endoresection in high posterior choroidal melanomas: Long-term outcome. *Br J Ophthalmol* 2008;92:1040-5.
51. García-Arumí J, Leila M, Zapata MA, Velázquez D, Dinarens-Fernandez MC, Tresserra F, *et al.* Endoresection technique with/without brachytherapy for management of high posterior choroidal melanoma: Extended follow-up results. *Retina* 2015;35:628-37.
52. Karkhanavaz R, Chams H, Amoli FA, Riazi-Esfahani M, Ahmadiabadi MN, Mansouri MR, *et al.* Long-term surgical outcome of posterior choroidal melanoma treated by endoresection. *Retina* 2007;27:908-14.
53. Süsskind D, Dürr C, Paulsen F, Kaulich T, Bartz-Schmidt KU. Endoresection with adjuvant ruthenium brachytherapy for selected uveal melanoma patients – The tuebingen experience. *Acta Ophthalmol* 2017;95:e727-33.
54. Ledowski T, Kiese F, Jeglin S, Scholz J. Possible air embolism during eye surgery. *Anesth Analg* 2005;100:1651-2.
55. Rice JC, Liebenberg L, Scholtz RP, Torr G. Fatal air embolism during endoresection of choroidal melanoma. *Retin Cases Brief Rep* 2014;8:127-9.
56. Morris RE, Sapp MR, Oltmanns MH, Kuhn F. Presumed air by vitrectomy embolisation (PAVE) a potentially fatal syndrome. *Br J Ophthalmol* 2014;98:765-8.
57. Morris RE, Boyd GL, Sapp MR, Oltmanns MH, Kuhn F, Albin MS. Ocular Venous Air Embolism (OVAE): A review. *J VitreoRetinal Diseases* 2019;3:99-106.
58. Huang Y, Wei WB. Acute pulmonary embolism caused by local resection of choroidal melanoma. *Chin Med J (Engl)* 2016;129:743-5.
59. Gayer S, Palte HD, Albini TA, Flynn HW Jr., Martinez-Ruiz R, Salas N, *et al.* *In vivo* porcine model of venous air embolism during pars plana vitrectomy. *Am J Ophthalmol* 2016;171:139-44.
60. Meraz Gutiérrez MP, Camara Rodriguez EJ, Pando Cifuentes A, Ortiz-Ramirez GY, Soberón Ventura V. Venous-air embolism during vitrectomy for endoresection of choroidal melanoma: Case report. *Eur J Ophthalmol* 2022;32:P173-6.
61. Ruschen H, Romano MR, Ferrara M, Loh GK, Wickham L, Damato BE, *et al.* Perfluorocarbon syndrome-a possible, overlooked source of fatal gas embolism following uveal-melanoma endoresection. *Eye (Lond)* 2022;36:2348-9.

62. Fiorentzis M, Bechrakis NE. Vortex vein cauterization and truncation to avoid perfluorocarbon syndrome during endoresection of uveal melanomas: A retrospective study. *Eye (Lond)* 2023;37:189-90.
63. Caminal JM, Mejia K, Masuet-Aumatell C, Arias L, Piulats JM, Gutierrez C, *et al.* Endoresection versus iodine-125 plaque brachytherapy for the treatment of choroidal melanoma. *Am J Ophthalmol* 2013;156:334-42.e1.
64. Mittica N, Vemuganti GK, Duffy M, Torczynski E, Edward DP. Late orbital recurrence of a choroidal melanoma following internal resection: Report of a case and review of the literature. *Surv Ophthalmol* 2003;48:181-90.
65. Robertson DM. Melanoma endoresection: A perspective. *Retina* 2001;21:403-7.
66. Onken MD, Worley LA, Ehlers JP, Harbour JW. Gene expression profiling in uveal melanoma reveals two molecular classes and predicts metastatic death. *Cancer Res* 2004;64:7205-9.
67. Damato B, Dopierala JA, Coupland SE. Genotypic profiling of 452 choroidal melanomas with multiplex ligation-dependent probe amplification. *Clin Cancer Res* 2010;16:6083-92.
68. Damato B. Choroidal melanoma endoresection, dandelions and allegory-based medicine. *Br J Ophthalmol* 2008;92:1013-4.
69. Hiong A, O'Day R, Fog LS, McKay D, McKenzie J, Ameratunga M, *et al.* Globe salvage and vision preservation by neoadjuvant darovasertib and crizotinib in uveal melanoma. *Ophthalmol Retina* 2024;8:325-30.